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California State University, Northridge

Department of Electrical & Computer Engineering

ECE 528

Kennel Bait

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Introduction

The main purpose for this project was to capture stray cats by designing and implementing embedded systems that implement a cage, servo, laser, and ultra sonic sensor. By knowing a cats major weakness the laser played a big role in trapping a cat in efficient and safe manner.

Kennel Bait Process

Both hardware and programming were being worked on simultaneously in order for an effective capture. Each component had to be in the correct placements. If the sensor was off the bait would have been ineffective. In addition, each delay needed a perfect timing so the cat would not have been startled.

***Figure 1**, Before Hardware*

Figure 1 shows the cage before the installation began. It took a few minutes on where the laser and servo would be mounted.

Figure 2 *– laser with servo installed*

The laser with the servo had to be mounted first (top right side of the cage). It needed to be placed at the right height in order for the laser to point from the outside. Once the placement was correct, the Pir motion sensor was installed near the servo with the laser. However, the Pir needed to be replaced with a ultra sonic sensor because the Pir motion sensor had a major delay in outputting the signal that notified the servo that closed the door.

Figure 3 *– Door preparation and installation*

The door had various tests for a proper open and closing behavior. The regular SG90 servo could not support the weight of the door. This led to a MG995 servo motor upgrade. This replacement can hold more weight and can move the door with ease.

Figure 4*, MG995 Servo Installation*

The MG995 servo is a better replacement. After coming up with the mount for the door. It was able to rotate the door with ease.

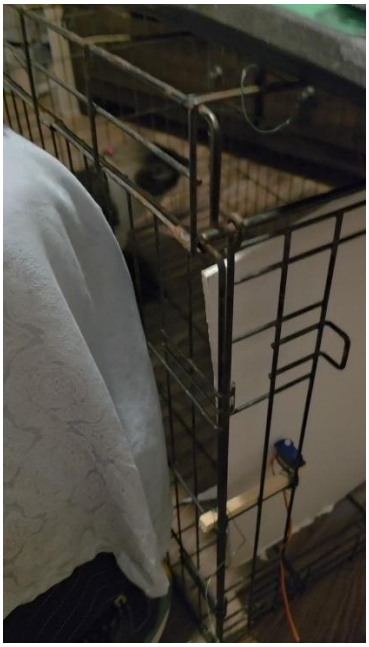

Figure 5- Demo cat capture

Figure 5 shows the cat getting captures once the components are installed. Once the sensor did not read detection. It waited 9 seconds and then opened back up to 0 degrees.

Conclusion

In conclusion, working on the kennel bait was a great experience. Not only do we apply what we learn in class but also have a creative mindset when solving problems and designing the system. A practical solution was applied by using the components such as a servo, laser, ultra sonic sensor, and a cage.